

Dual Trigger

The data except this specification conform to those of each product.

1. Product type

IMX530 series

IMX530-AAMJ-C, IMX530-AAQJ-C, IMX531-AAMJ-C, IMX531-AAQJ-C, IMX532-AAMJ-C, IMX532-AAQJ-C

IMX535 series

IMX535-AAMJ-C, IMX535-AAQJ-C, IMX536-AAMJ-C, IMX536-AAQJ-C, IMX537-AAMJ-C, IMX537-AAQJ-C

IMX540 series

IMX540-AAMJ-C, IMX540-AAQJ-C, IMX541-AAMJ-C, IMX541-AAQJ-C, IMX542-AAMJ-C, IMX542-AAQJ-C

IMX545 series

IMX545-AAMJ-C, IMX545-AAQJ-C, IMX546-AAMJ-C, IMX546-AAQJ-C, IMX547-AAMJ-C, IMX547-AAQJ-C

IMX548 series

IMX548-AAMJ-C, IMX548-AAQJ-C

IMX565 series

IMX565-AAMJ-C, IMX565-AAQJ-C, IMX566-AAMJ-C, IMX566-AAQJ-C, IMX567-AAMJ-C, IMX567-AAQJ-C

IMX568 series

IMX568-AAMJ-C, IMX568-AAQJ-C

2. Description

The function of Dual Trigger is described. This function can divide the pixel area into 2 areas and can set the different exposure time to each area. And in case of using Overlap ROI, this function can set the different exposure time and gain to each area.

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3. Features

- ◆ Dual Trigger function
 - Switching Exposure Time setting
 - Switching Gain setting
 - * Vertical Inverted Operation during use of this function doesn't support.
- ◆ Support mode
 - Sequential Trigger mode
 - Fast Trigger mode
- ◆ Support readout drive mode
 - Switching Exposure Time:
 - All-pixel scan mode
 - Overlap ROI mode (ROI mode is not supported.)
 - Switching Gain:
 - Overlap ROI mode (ROI mode is not supported.)
- ◆ The monitor output for Exposure period
- ◆ Support interface

(IMX530 series / IMX535 series)

Interface
SLVS – EC output (1 lane / 2 lane / 4 lane / 8 lane, 8 bit / 10 bit / 12 bit output)
SLVS output (4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX540 series / IMX545 series)

Interface
SLVS – EC output (1 lane / 2 lane, 8 bit / 10 bit / 12 bit output)
SLVS output (2 ch / 4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX548 series)

Interface
SLVS output (2 ch / 4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX565 series / IMX568 series)

Interface
CSI-2 serial data output (2 Lane / 4 Lane, 8 bit / 10 bit / 12 bit output)

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4. Register Map

The register map for Dual Trigger is shown below.

Please refer to the product specification for register setup other than those list.

Registers corresponding to Chip ID = 02h in Write mode. (I²C:30**h)

Address		bit	Register Name	Description	Default value after reset				Reflection timing		
4-wire	I ² C				By register		By address				
E5h	30E5h	0	GAINDLY [7:0]	Gain switching wait time setting Refer to the register list in each Readout mode in the data sheet (0h: Setting prohibited)	04h ^{*1}	02h ^{*2}	04h ^{*1}	02h ^{*2}	S		
		1									
		2									
		3									
		4									
		5									
		6									
E6h	30E6h	0	GSDLY [7:0]	Refer to the register list in each Readout mode in the data sheet (0h: Setting prohibited)	34h ^{*3}	30h ^{*4}	12h ^{*5}	34h ^{*3}	30h ^{*4}	12h ^{*5}	S
		1									
		2									
		3									
		4									
		5									
		6									
7											

^{*1} IMX530 series, IMX535 series, IMX540 series, IMX545 series and IMX565 series.

^{*2} IMX548 series and IMX568 series.

^{*3} IMX530 series and IMX540 series.

^{*4} IMX535 series, IMX545 series and IMX565 series.

^{*5} IMX548 series and IMX568 series.

Registers corresponding to Chip ID = 04h in Write mode. (I²C:32**h)

Address		bit	Register Name	Description	Default value after reset		Reflection timing			
4-wire	I ² C				By register	By address				
50h	3250h	0	AREAADD [2:0]	Dual trigger area setting (v direction) 0h: Upper_area (XTRIG1) = all area 1h: Upper_area (XTRIG1) = AREA 1 to 7 Lower_area (XTRIG2) = AREA 8 2h: Upper_area (XTRIG1) = AREA 1 to 6 Lower_area (XTRIG2) = AREA 7 to 8 3h: Upper_area (XTRIG1) = AREA 1 to 5 Lower_area (XTRIG2) = AREA 6 to 8 4h: Upper_area (XTRIG1) = AREA 1 to 4 Lower_area (XTRIG2) = AREA 5 to 8 5h: Upper_area (XTRIG1) = AREA 1 to 3 Lower_area (XTRIG2) = AREA 4 to 8 6h: Upper_area (XTRIG1) = AREA 1 to 2 Lower_area (XTRIG2) = AREA 3 to 8 7h: Upper_area (XTRIG1) = AREA 1 Lower_area (XTRIG2) = AREA 2 to 8	0h	00h	S			
		1								
		2								
		3						Fixed to 0	0	—
		4						Fixed to 0	0	—
		5						Fixed to 0	0	—
		6						Fixed to 0	0	—
		7						Fixed to 0	0	—

Registers corresponding to Chip ID = 06h in Write mode. (I²C:34**h)

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
00h	3400h	0	TRIGMODE [2:0]	Global shutter mode setting 0h: Normal mode 1h: Sequential Trigger mode 2h: Fast Trigger mode Others: Setting prohibited	0h	00h	S
		1					
		2					
		3	TRIGTIMING [1:0]	Trigger mode setting 0h: Normal mode 2h: Dual trigger mode Others: Setting prohibited	0h		S
		4					
		5		Fixed to 0			
		6	Fixed to 0	0	—		
		7	Fixed to 0	0	—		
36h	3436h	0	TOUT0SEL [2:0]	TOUT0 pin setting 0h: Low fixed 1h: XTRIG1 Exposure period monitoring 2h: XTRIG2 Exposure period monitoring	0h	00h	S
		1					
		2					
		3	TOUT1EXPSEL [2:0]	TOUT1 pin setting 0h: Low fixed 1h: XTRIG1 Exposure period monitoring 2h: XTRIG2 Exposure period monitoring	0h		S
		4					
		5					
		6	TOUT2EXPSEL [2:0]	TOUT2 pin setting 0h: Low fixed 1h: XTRIG1 Exposure period monitoring 2h: XTRIG2 Exposure period monitoring	0h		S
		7					
37h	3437h	0				00h	
		1	Fixed to 0	0	—		
		2	Fixed to 0	0	—		
		3	Fixed to 0	0	—		
		4	Fixed to 0	0	—		
		5	Fixed to 0	0	—		
		6	Fixed to 0	0	—		
		7	Fixed to 0	0	—		

Registers corresponding to Chip ID = 07h in Write mode. (I²C:35**h)

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
03h	3503h	0	AREAGAIN_EN [0]	Switching Gain Enable 0: Switching Gain Off 1: Switching Gain On	0	00h	S
		1		Fixed to 0	0		—
		2		Fixed to 0	0		—
		3		Fixed to 0	0		—
		4	AREAGAIN_ROISEL [2:0]	Dividing gain area setting (v direction) 0h: GAIN is divided by end of ROI1 area 1h: GAIN is divided by end of ROI2 area 2h: GAIN is divided by end of ROI3 area	0h		S
		5		3h: GAIN is divided by end of ROI4 area 4h: GAIN is divided by end of ROI5 area 5h: GAIN is divided by end of ROI6 area			
		6		6h: GAIN is divided by end of ROI7 area 7h: GAIN is divided by end of ROI8 area			
		7		Fixed to 0			

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Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
14h	3514h	0	GAIN [8:0]	LSB	000h	00h	V
		1		Upper area GAIN setting			
		2					
		3					
		4					
		5					
		6					
		7					
15h	3515h	0		MSB	0	00h	—
		1		Fixed to 0			
		2		Fixed to 0			
		3		Fixed to 0			
		4		Fixed to 0			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
16h	3516h	0	GAIN1 [8:0]	LSB	000h	00h	V
		1		Lower area GAIN setting			
		2					
		3					
		4					
		5					
		6					
		7					
17h	3517h	0		MSB	0	00h	—
		1		Fixed to 0			
		2		Fixed to 0			
		3		Fixed to 0			
		4		Fixed to 0			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			

5. Dual Trigger

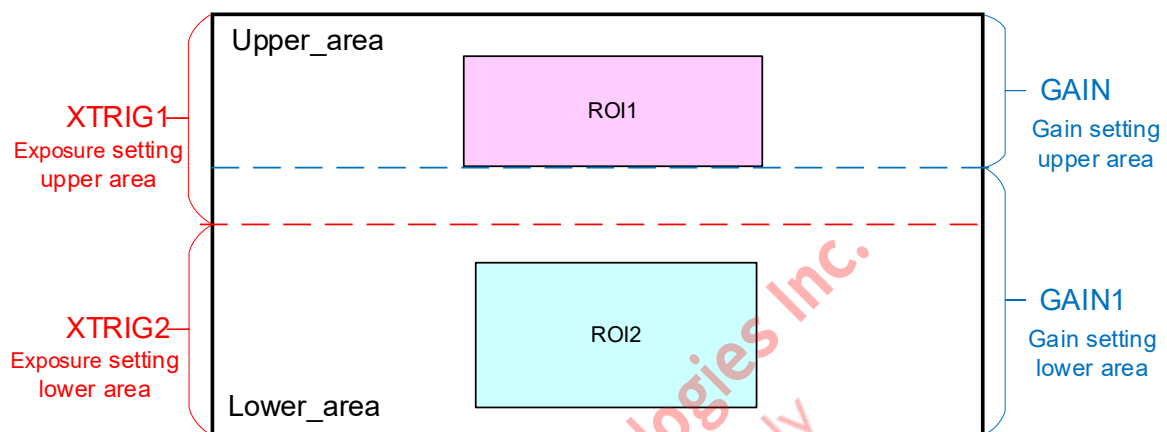
5.1. Overview

Divide the screen into two areas, and output images with different exposure times / gains in each area.
* It is also possible to cropping arbitrary area using Overlap ROI within each 2 areas.

Dual trigger has the following two functions.

- (1) Switching Exposure Time setting
- (2) Switching Gain setting

These two functions can be set together.



Dual Trigger Image

6. Switching Exposure Time Setting

This function can divide the effective pixels into 2 areas, and set the each exposure timing to Upper area and Lower area.

This function supports All-pixel scan mode and Overlap ROI mode.

6.1. Exposure Time Setting

The each exposure time of Upper area and Lower area is set by inputting signal to external trigger pin (XTRIG1, XTRIG2).

The exposure time of each area is defined as below.

Upper area: XTRIG1

Lower area: XTRIG2

The exposure time of each area is started by fall input of XTRIG1 and XTRIG2, and the exposure time of both areas is ended by rise input of XTRIG1. About the exposure timing, refer to section of "Dual Trigger Operation".

6.2. Switching Exposure Timing Setting

Set registers as below

Chip ID	Address	Bit	register name	Note
02h	E5h	[7:0]	GAINDLY	Refer to each driving mode of datasheet
	E6h	[7:0]	GSDLY	Refer to each driving mode of datasheet
04h	50h	[2:0]	AREAADD	Set the Dual trigger area
06h	00h	[2:0]	TRIGMODE	Global shutter mode setting 1h: Sequential Trigger mode 2h: Fast Trigger mode
		[4:3]	TRIGTIMING	2h: Dual trigger

6.3. Exposure Setting Area and Division Setting

The pixel area is divided shown as below.

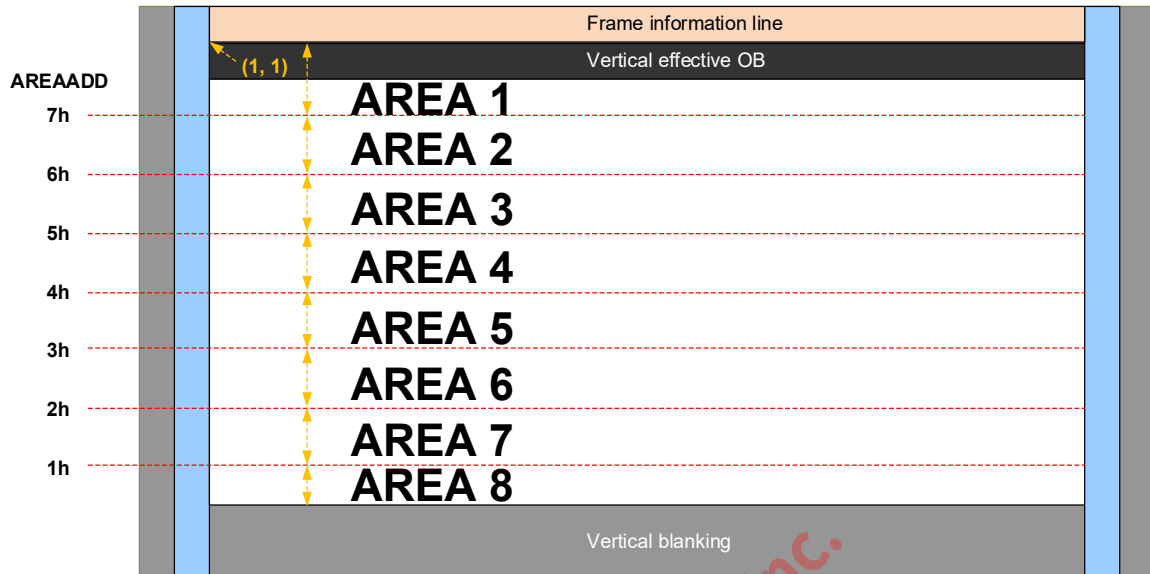


Image of Exposure Setting Area

Area	IMX530-AAMJ-C IMX530-AAQJ-C IMX540-AAMJ-C IMX540-AAQJ-C		IMX531-AAMJ-C IMX531-AAQJ-C IMX541-AAMJ-C IMX541-AAQJ-C		IMX532-AAMJ-C IMX532-AAQJ-C IMX542-AAMJ-C IMX542-AAQJ-C	
	Start Line	End Line	Start Line	End Line	Start Line	End Line
1	1	640	1	592	1	64
2	641	1216	593	1168	65	432
3	1217	1792	1169	1744	433	1008
4	1793	2368	1745	2320	1009	1584
5	2369	2944	2321	2896	1585	2160
6	2945	3520	2897	3472	2161	2736
7	3521	4096	3473	4048	2737	3104
8	4097	4672	4049	4576	none	

Area	IMX535-AAMJ-C IMX535-AAQJ-C IMX545-AAMJ-C IMX545-AAQJ-C IMX565-AAMJ-C IMX565-AAQJ-C		IMX536-AAMJ-C IMX536-AAQJ-C IMX546-AAMJ-C IMX546-AAQJ-C IMX566-AAMJ-C IMX566-AAQJ-C		IMX537-AAMJ-C IMX537-AAQJ-C IMX547-AAMJ-C IMX547-AAQJ-C IMX567-AAMJ-C IMX567-AAQJ-C	
	Start Line	End Line	Start Line	End Line	Start Line	End Line
1	1	440	1	360	1	64
2	441	816	361	736	65	344
3	817	1192	737	1112	345	720
4	1193	1568	1113	1488	721	1096
5	1569	1944	1489	1864	1097	1472
6	1945	2320	1865	2240	1473	1848
7	2321	2696	2241	2616	1849	2128
8	2697	3072	2617	2912	none	

Area	IMX548-AAMJ-C IMX548-AAQJ-C IMX568-AAMJ-C IMX568-AAQJ-C	
	Start Line	End Line
1	1	322
2	323	580
3	581	838
4	839	1096
5	1097	1354
6	1355	1612
7	1613	1870
8	1871	2128

Upper and Lower area of Exposure time setting is set by AREAADD register (Chip ID = 04h, Address = 50h). It sets the area where is divided to Upper and Lower area.

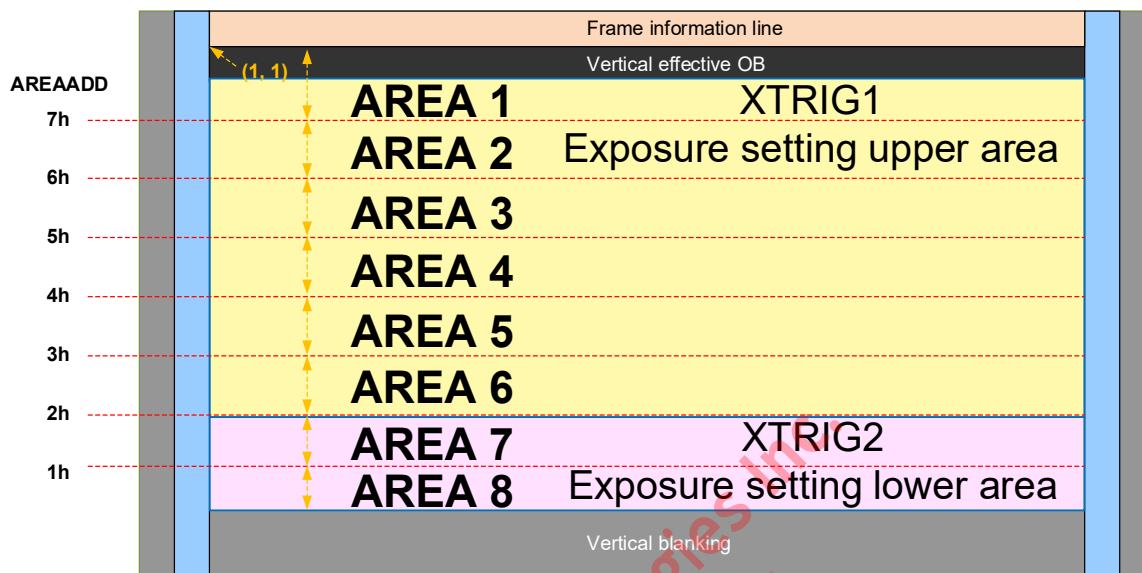
Setting	Exposure setting upper area (XTRIG1)	Exposure setting lower area (XTRIG2)
0h	All AREA	none
1h	AREA 1 to AREA 7	AREA 8
2h	AREA 1 to AREA 6	AREA 7, AREA 8
3h	AREA 1 to AREA 5	AREA 6 to AREA 8
4h	AREA 1 to AREA 4	AREA 5 to AREA 8
5h	AREA 1 to AREA 3	AREA 4 to AREA 8
6h	AREA 1, AREA 2	AREA 3 to AREA 8
7h	AREA 1	AREA 2 to AREA 8

6.3.1. The Example of Exposure Time Division Setting

All-pixel scan

The example of division setting is shown below

AREAADD = 2h, SLVS output

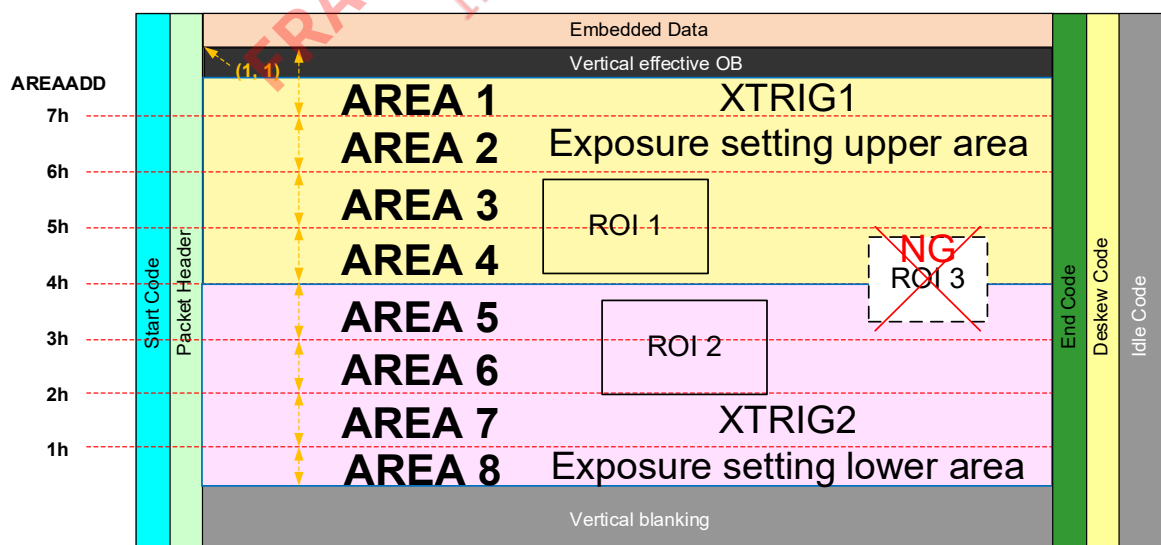


The example of area divide setting for the exposure (All-pixel scan, SLVS output)

Overlap ROI mode

About setting and restriction of Overlap ROI, refer to Overlap ROI mode on the datasheet. Avoid ROI area setting on the division line of exposure.

AREAADD = 4h, Overlap ROI mode, SLVS – EC output



The example of area divide setting for the exposure (Overlap ROI mode, SLVS – EC output)

7. Switching Gain Setting

This function can set 2 areas divided by the end of selected ROI area line, and set each gain to Upper area and Lower area.

This function supports only Overlap ROI mode.

7.1. Gain Setting

The each gain of Upper area and Lower area is set as below.

Upper area: GAIN (ChipID = 07h, Address = 14h, 15h)

Lower area: GAIN1 (ChipID = 07h, Address = 16h, 17h)

7.2. Switching Gain Setting

Set registers as below.

Chip ID	Address	Bit	register name	Note
07h	03h	[0]	AREAGAIN_EN	1: Switching Gain On
		[6:4]	AREAGAIN_ROISEL	Set the switching gain area
	14h	[7:0]	GAIN	Set Upper area gain
	15h	[1:0]		
	16h	[7:0]	GAIN1	Set Lower area gain
	17h	[1:0]		

7.3. Gain setting Area and Division Setting

The division area is set by AREAGAIN_ROISEL register (ChipID = 07h, Address = 03h).

The gain is switched by the end of line of the selected ROI area.

Set ROI area before this setting.

The divided line between Upper area and Lower area is changed by the ROI area setting.

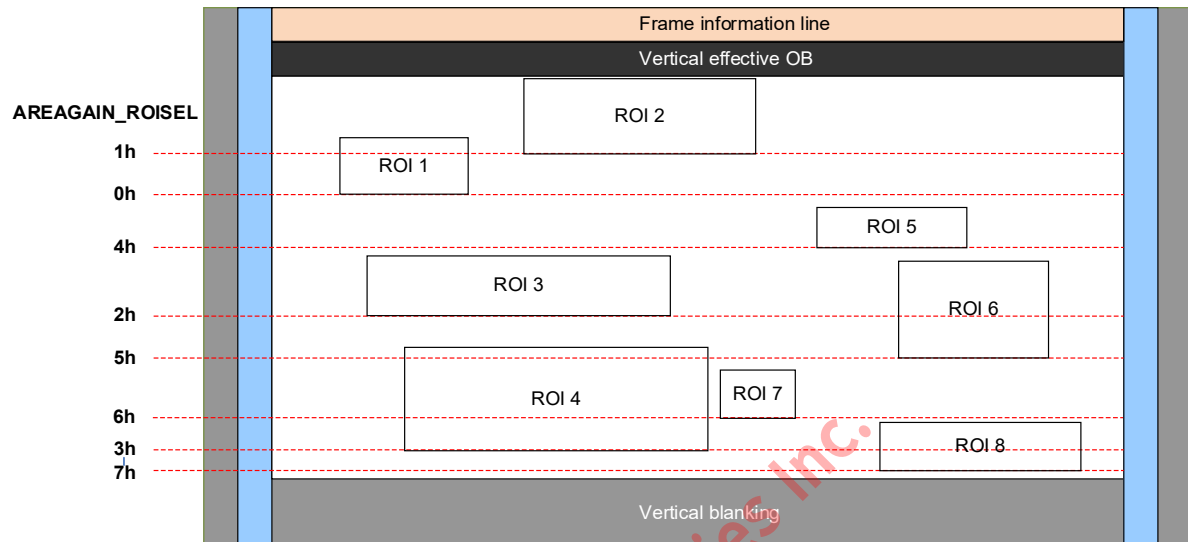


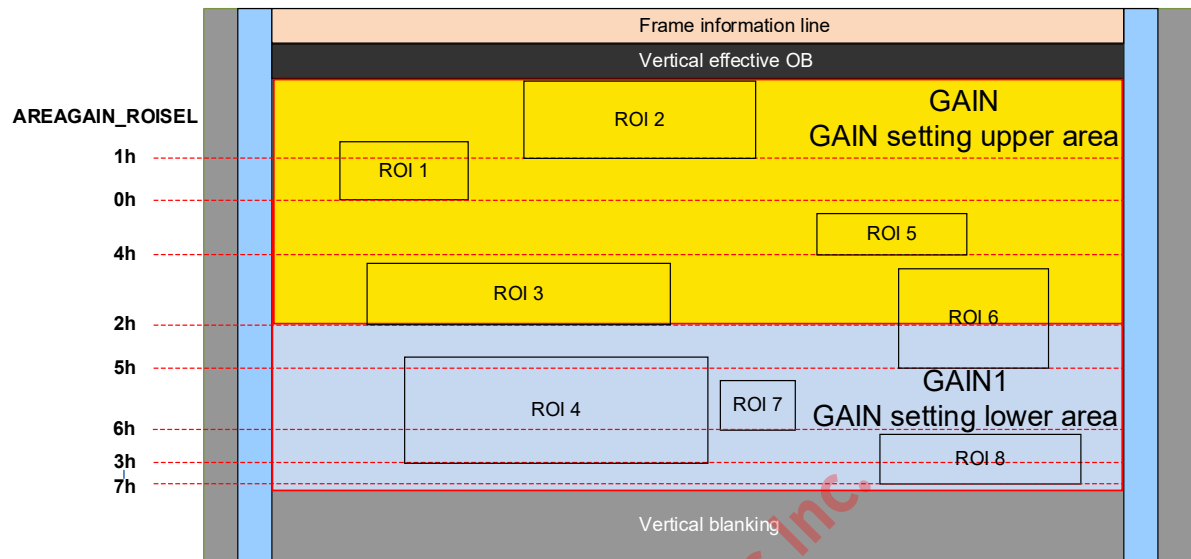
Image of Gain Setting Area

- * If the ROI area selected by the AREAGAIN_ROISEL register was the last area (select ROI8 area in the above figure), gain switching is not performed. Gain switching wait time also does not occur.

7.3.1. The Example of Gain Division setting

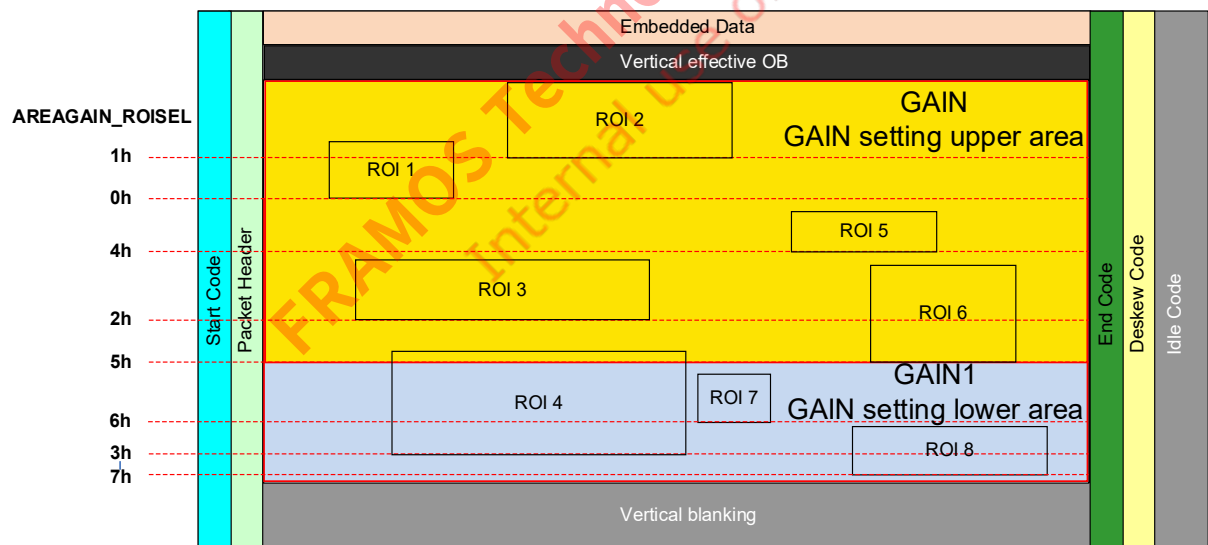
The example of division setting is shown below.

AREAADD = 2h, SLVS output



The example of area divide setting for the gain (SLVS output)

AREAADD = 5h, SLVS-EC output



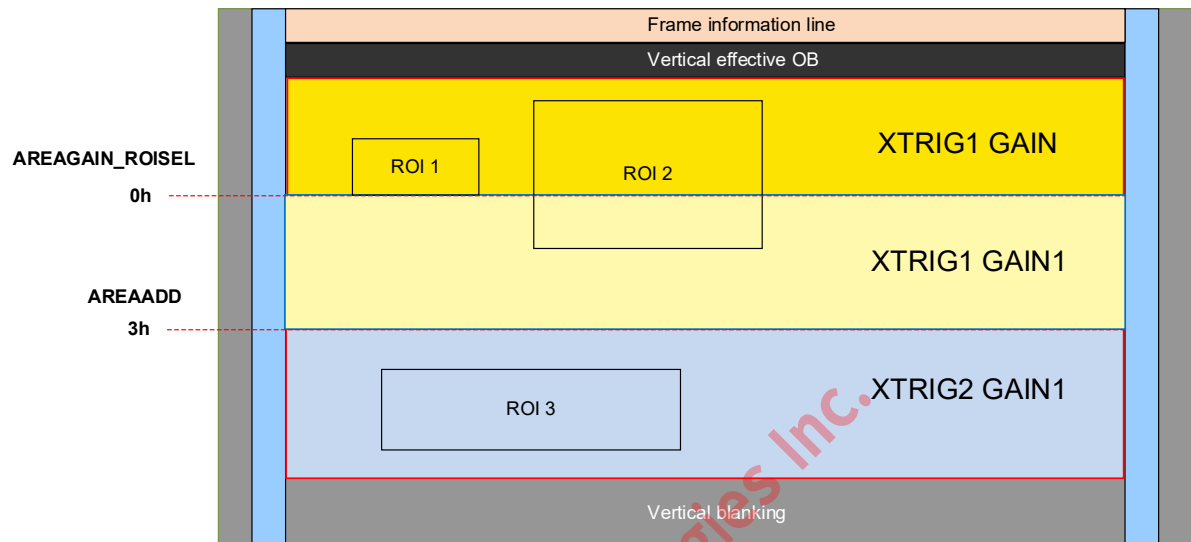
The example of area divide setting for the gain (SLVS output)

8. Example of setting the both Switching Exposure Time and Switching Gain

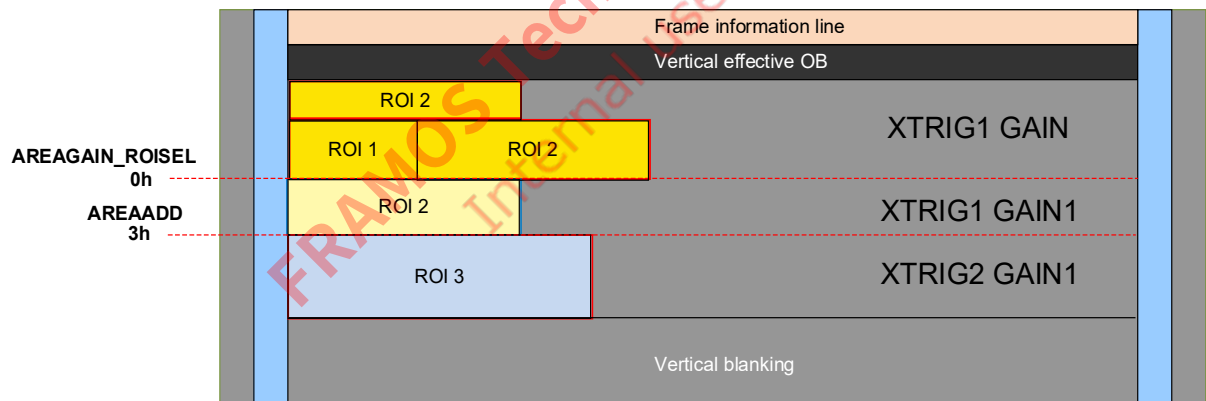
The example of divide setting both exposure time and gain is as shown as below.

AREAADD = 3h, AREAGAIN_ROISEL = 0h, ROI Overlap mode, SLVS output

Exposure Time division and Gain division must not be corresponded because of setting.



The example of the setting image of area setting for the exposure and the gain



The example of the output image of area setting for the exposure and the gain

9. Dual Trigger Operation

This function supports Sequential trigger mode and Fast trigger mode by inputting external trigger.
Set the trigger mode by the TRIGMODE register.

Start timing of the exposure is as below.

Upper area: Fall input of XTRIG1

Lower area: Fall input of XTRIG2

End timing of the exposure is as below.

Both area: Rise input of XTRIG1

(Rise input of XTRIG2 doesn't affect the end of the exposure and it is ignored.)

Set XTRIG2 into High after the exposure time and before the start of next exposure.

The shutter stabilization time is occurred like normal mode.

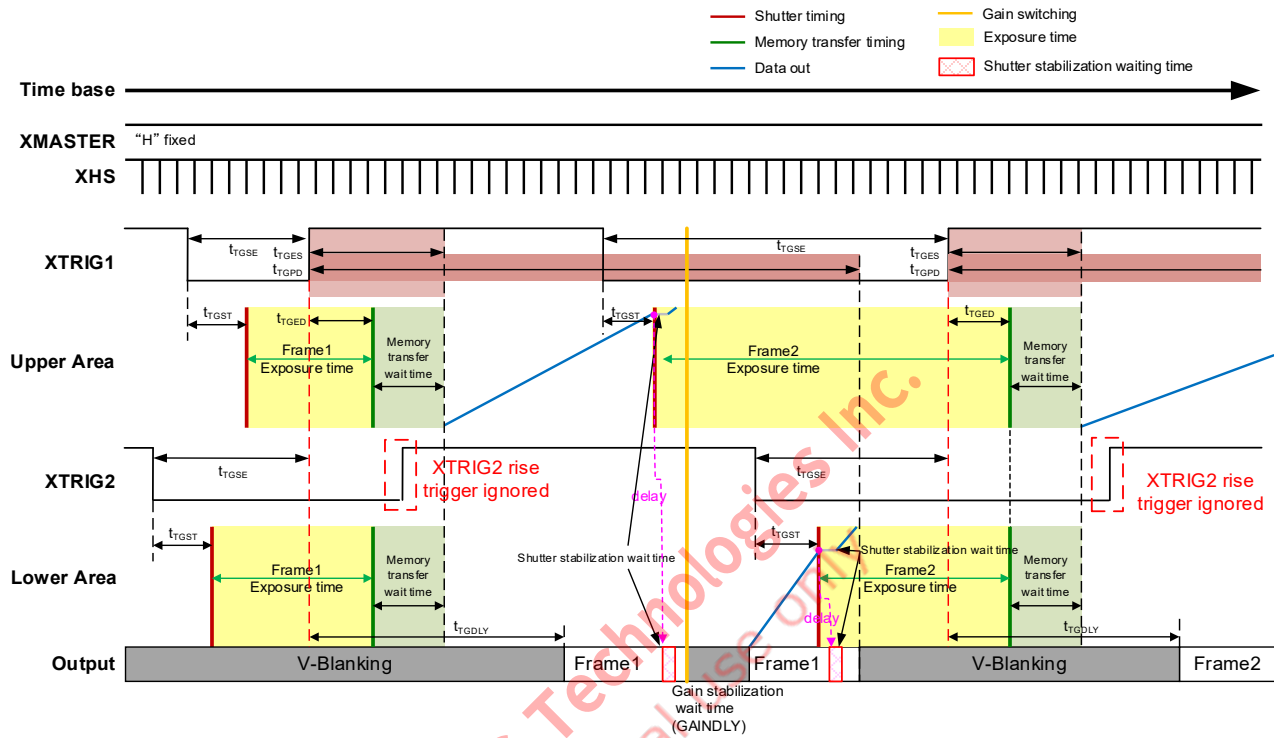
In case of selecting Gain Switching, the gain stabilization time is occurred at the timing of switching gain in the frame and blank is output. Gain Stabilization time is defined by GAINDLY.

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9.1. Dual Trigger (Sequential Trigger Mode)

Exposure time is controlled by XTRIG1 / XTRIG2 signals. In case of inputting continually, there are period which prohibit the trigger rise input (t_{TGPD}) and fall input (t_{TGES}) based on the previous trigger rise.

Next rise input of XTRIG1 can't be input until the end of read out of the previous rise input of XTRIG1. (It is possible to input the fall signal into XTRIG1 / XTRIG2.) The shutter stabilization wait time is occurred in upper and lower areas, it depends on exposure time setting.



Sequential trigger mode operation timing

9.1.1. Settings

Parameter List of Sequential Trigger Mode

(IMX530 series / IMX535-AAQJ-C / IMX536-AAQJ-C / IMX537-AAQJ-C)

Item	Symbol	Min.	Typ.	Max.	Unit
Integration start delay	t_{TGST}	$4 + GMRWT^{*1}$	—	$5 + GMRWT^{*1}$	H
Integration end delay	t_{TGED}	$4 + GMRWT^{*1} + t_{OFFSET}$	—	$5 + GMRWT^{*1} + t_{OFFSET}$	H
Pulse width	t_{TGSE}	1	—	—	H
Next trigger fall prohibited period	t_{TGES}	$5 + GMRWT^{*1} + GMTWT^{*1}$	—	—	H
Next trigger rise prohibited period (All-pixel)	t_{TGPD}	V _{MAX}	—	—	H
Next trigger rise prohibited period (ROI)		V_{TR}^{*2}	—	—	H
Data output delay	t_{TGDLY}	—	$25 + GMRWT^{*1} + GMTWT^{*1}$	—	H

*1 For GMRWT and GMTWT, refer to the register list of each readout mode.

*2 Overlap ROI: V_{TR} = The number of vertical lines

(IMX540 series / IMX545 series / IMX548 series / IMX565 series / IMX568 series)

Item	Symbol	Min.	Typ.	Max.	Unit
Integration start delay	t_{TGST}	$4 + GMRWT^{*1}$	—	$5 + GMRWT^{*1}$	H
Integration end delay	t_{TGED}	$4 + GMRWT^{*1} + t_{OFFSET}$	—	$5 + GMRWT^{*1} + t_{OFFSET}$	H
Pulse width	t_{TGSE}	1	—	—	H
Next trigger fall prohibited period	t_{TGES}	$5 + GMRWT^{*1} + GMTWT^{*1}$	—	—	H
Next trigger rise prohibited period (All-pixel)	t_{TGPD}	V _{MAX}	—	—	H
Next trigger rise prohibited period (ROI)		V_{TR}^{*2}	—	—	H
Data output delay	t_{TGDLY}	—	$15 + GMRWT^{*1} + GMTWT^{*1}$	—	H

*1 For GMRWT and GMTWT, refer to the register list of each readout mode.

*2 Overlap ROI: V_{TR} = The number of vertical lines

(IMX535-AAMJ-C / IMX536-AAMJ-C / IMX537-AAMJ-C)

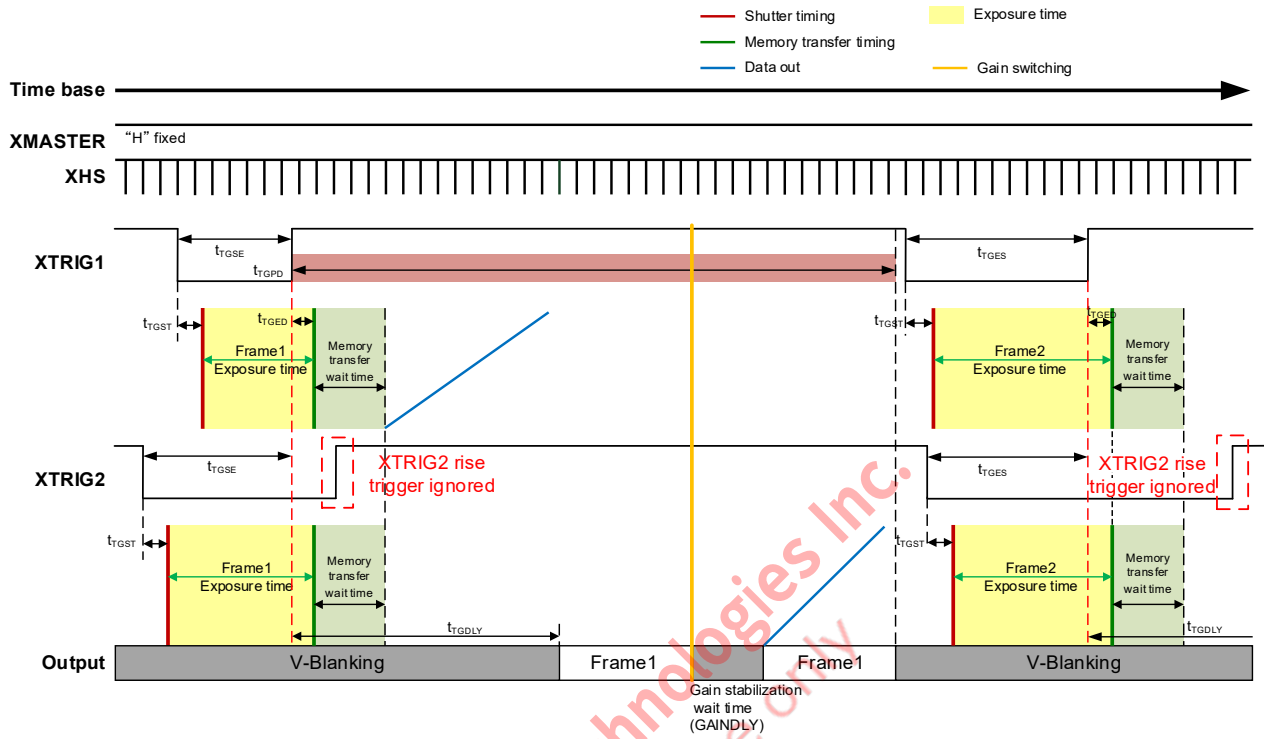
Item	Symbol	Min.	Typ.	Max.	Unit
Integration start delay	t_{TGST}	$8 + GMRWT^{*1}$	—	$9 + GMRWT^{*1}$	H
Integration end delay	t_{TGED}	$8 + GMRWT^{*1} + t_{OFFSET}$	—	$9 + GMRWT^{*1} + t_{OFFSET}$	H
Pulse width	t_{TGSE}	1	—	—	H
Next trigger fall prohibited period	t_{TGES}	$9 + GMRWT^{*1} + GMTWT^{*1}$	—	—	H
Next trigger rise prohibited period (All-pixel)	t_{TGPD}	V _{MAX}	—	—	H
Next trigger rise prohibited period (ROI)		V_{TR}^{*2}	—	—	H
Data output delay	t_{TGDLY}	—	$29 + GMRWT^{*1} + GMTWT^{*1}$	—	H

*1 For GMRWT and GMTWT, refer to the register list of each readout mode.

*2 Overlap ROI: V_{TR} = The number of vertical lines

9.2. Dual Trigger (Sequential Trigger Mode) (Use gain switching)

Exposure time is controlled by XTRIG1 / XTRIG2 signals. It is impossible to start the next exposure until the previous read out finished after the last rise input of XTRIG1.



Sequential trigger mode operation timing(Use gain switching)

9.2.1. Settings

Parameter List of Sequential Trigger Mode

(IMX530 series / IMX535-AAQJ-C / IMX536-AAQJ-C / IMX537-AAQJ-C)

Item	Symbol	Min.	Typ.	Max.	Unit
Integration start delay	t_{TGST}	$4 + GMRWT^{*1}$	—	$5 + GMRWT^{*1}$	H
Integration end delay	t_{TGED}	$4 + GMRWT^{*1} + t_{OFFSET}$	—	$5 + GMRWT^{*1} + t_{OFFSET}$	H
Pulse width	t_{TGSE}	1	—	—	H
Next trigger rise / fall prohibited period (All-pixel)	t_{TGPD}	VMAX	—	—	H
Next trigger rise / fall prohibited period (ROI)		V_{TR}^{*2}	—	—	H
Data output delay	t_{TGDLY}	—	$25 + GMRWT^{*1} + GMTWT^{*1}$	—	H

*1 For GMRWT and GMTWT, refer to the register list of each readout mode.

*2 Overlap ROI: V_{TR} = The number of vertical lines

(IMX540 series / IMX545 series / IMX548 series / IMX565 series / IMX568 series)

Item	Symbol	Min.	Typ.	Max.	Unit
Integration start delay	t_{TGST}	$4 + GMRWT^{*1}$	—	$5 + GMRWT^{*1}$	H
Integration end delay	t_{TGED}	$4 + GMRWT^{*1} + t_{OFFSET}$	—	$5 + GMRWT^{*1} + t_{OFFSET}$	H
Pulse width	t_{TGSE}	1	—	—	H
Next trigger rise / fall prohibited period (All-pixel)	t_{TGPD}	VMAX	—	—	H
Next trigger rise / fall prohibited period (ROI)		V_{TR}^{*2}	—	—	H
Data output delay	t_{TGDLY}	—	$15 + GMRWT^{*1} + GMTWT^{*1}$	—	H

*1 For GMRWT and GMTWT, refer to the register list of each readout mode.

*2 Overlap ROI: V_{TR} = The number of vertical lines

(IMX535-AAMJ-C / IMX536-AAMJ-C / IMX537-AAMJ-C)

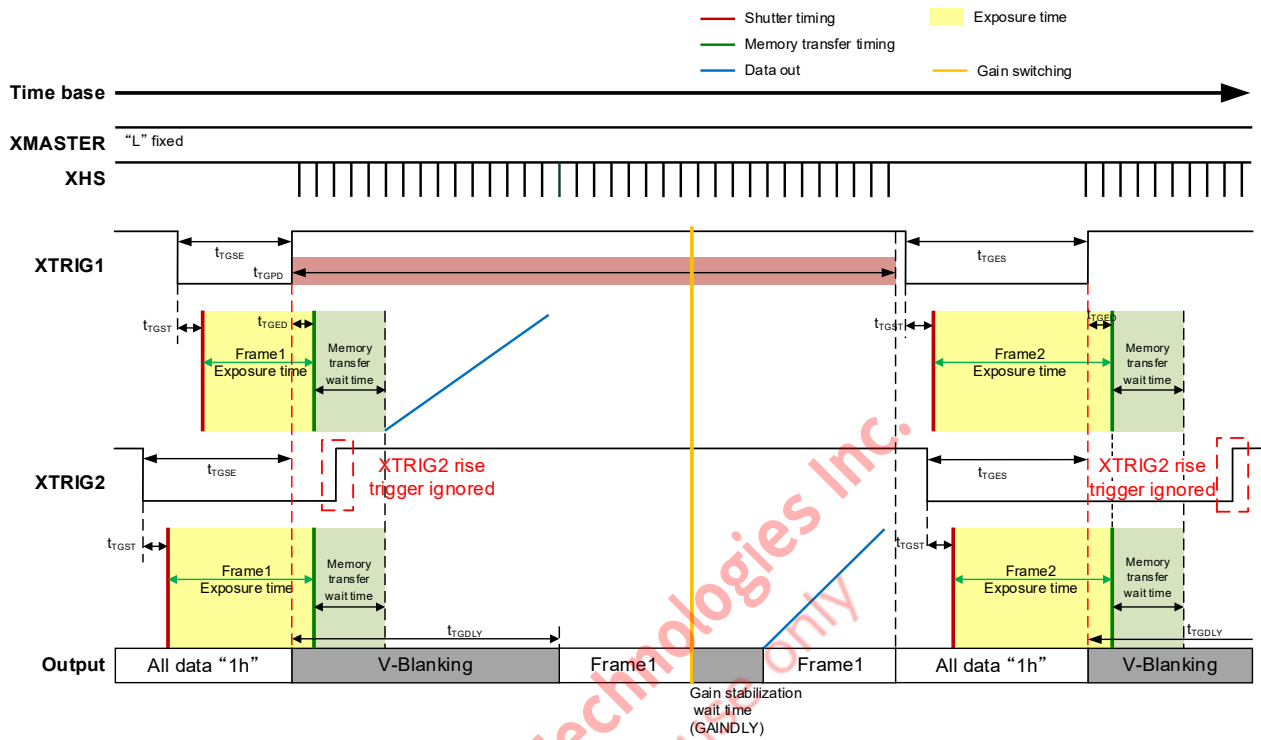
Item	Symbol	Min.	Typ.	Max.	Unit
Integration start delay	t_{TGST}	$8 + GMRWT^{*1}$	—	$9 + GMRWT^{*1}$	H
Integration end delay	t_{TGED}	$8 + GMRWT^{*1} + t_{OFFSET}$	—	$9 + GMRWT^{*1} + t_{OFFSET}$	H
Pulse width	t_{TGSE}	1	—	—	H
Next trigger rise / fall prohibited period (All-pixel)	t_{TGPD}	VMAX	—	—	H
Next trigger rise / fall prohibited period (ROI)		V_{TR}^{*2}	—	—	H
Data output delay	t_{TGDLY}	—	$29 + GMRWT^{*1} + GMTWT^{*1}$	—	H

*1 For GMRWT and GMTWT, refer to the register list of each readout mode.

*2 Overlap ROI: V_{TR} = The number of vertical lines

9.3. Dual trigger (Fast trigger mode)

Exposure time is controlled by XTRIG1 / XTRIG2 signals. It is impossible to start the next exposure until the previous read out finished after the last rise input of XTRIG1.



9.3.1. Settings

Parameter List of Fast Trigger Mode

(IMX530 series)

Item	Symbol	Min.	Typ.	Max.	Unit
Integration start delay	t_{TGST}	—	—	0.1	μs
Integration end delay	t_{TGED}	—	—	$0.1 + GMRWT^{*1} + t_{OFFSET}$	μs
Pulse width	t_{TGSE}	0.05	—	—	μs
Next trigger rise / fall prohibited period (All-pixel)	t_{TGPD}	$17 + V_{MAX}$	—	—	H
Next trigger rise / fall prohibited period (ROI)		$17 + V_{TR}^{*2}$	—	—	H
Data output delay	t_{TGDLY}	—	$21 + GMRWT^{*1} + GMTWT^{*1}$	—	H

^{*1} For GMRWT and GMTWT, refer to the register list of each readout mode.

^{*2} Overlap ROI: V_{TR} = The number of vertical lines

(IMX540 series)

Item	Symbol	Min.	Typ.	Max.	Unit
Integration start delay	t_{TGST}	—	—	0.1	μs
Integration end delay	t_{TGED}	—	—	$0.1 + GMRWT^{*1} + t_{OFFSET}$	μs
Pulse width	t_{TGSE}	0.05	—	—	μs
Next trigger rise / fall prohibited period (All-pixel)	t_{TGPD}	$7 + V_{MAX}$	—	—	H
Next trigger rise / fall prohibited period (ROI)		$7 + V_{TR}^{*2}$	—	—	H
Data output delay	t_{TGDLY}	—	$11 + GMRWT^{*1} + GMTWT^{*1}$	—	H

^{*1} For GMRWT and GMTWT, refer to the register list of each readout mode.

^{*2} Overlap ROI: V_{TR} = The number of vertical lines

(IMX535 series)

Item	Symbol	Min.	Typ.	Max.	Unit
Integration start delay	t_{TGST}	—	—	0.1	μs
Integration end delay	t_{TGED}	—	—	$0.1 + GMRWT^{*1} + t_{OFFSET}$	μs
Pulse width	t_{TGSE}	0.05	—	—	μs
Next trigger rise / fall prohibited period (All-pixel)	t_{TGPD}	V_{MAX}	—	—	H
Next trigger rise / fall prohibited period (ROI)		V_{TR}^{*2}	—	—	H
Data output delay	t_{TGDLY}	—	$21 + GMRWT^{*1} + GMTWT^{*1}$	—	H

^{*1} For GMRWT and GMTWT, refer to the register list of each readout mode.

^{*2} Overlap ROI: V_{TR} = The number of vertical lines

(IMX545 series / IMX548 series / IMX565 series / IMX568 series)

Item	Symbol	Min.	Typ.	Max.	Unit
Integration start delay	t_{TGST}	—	—	0.1	μs
Integration end delay	t_{TGED}	—	—	$0.1 + GMRWT^{*1} + t_{OFFSET}$	μs
Pulse width	t_{TGSE}	0.05	—	—	μs
Next trigger rise / fall prohibited period (All-pixel)	t_{TGPD}	V_{MAX}	—	—	H
Next trigger rise / fall prohibited period (ROI)		V_{TR}^{*2}	—	—	H
Data output delay	t_{TGDLY}	—	$11 + GMRWT^{*1} + GMTWT^{*1}$	—	H

*¹ For GMRWT and GMTWT, refer to the register list of each readout mode.*² Overlap ROI: V_{TR} = The number of vertical lines

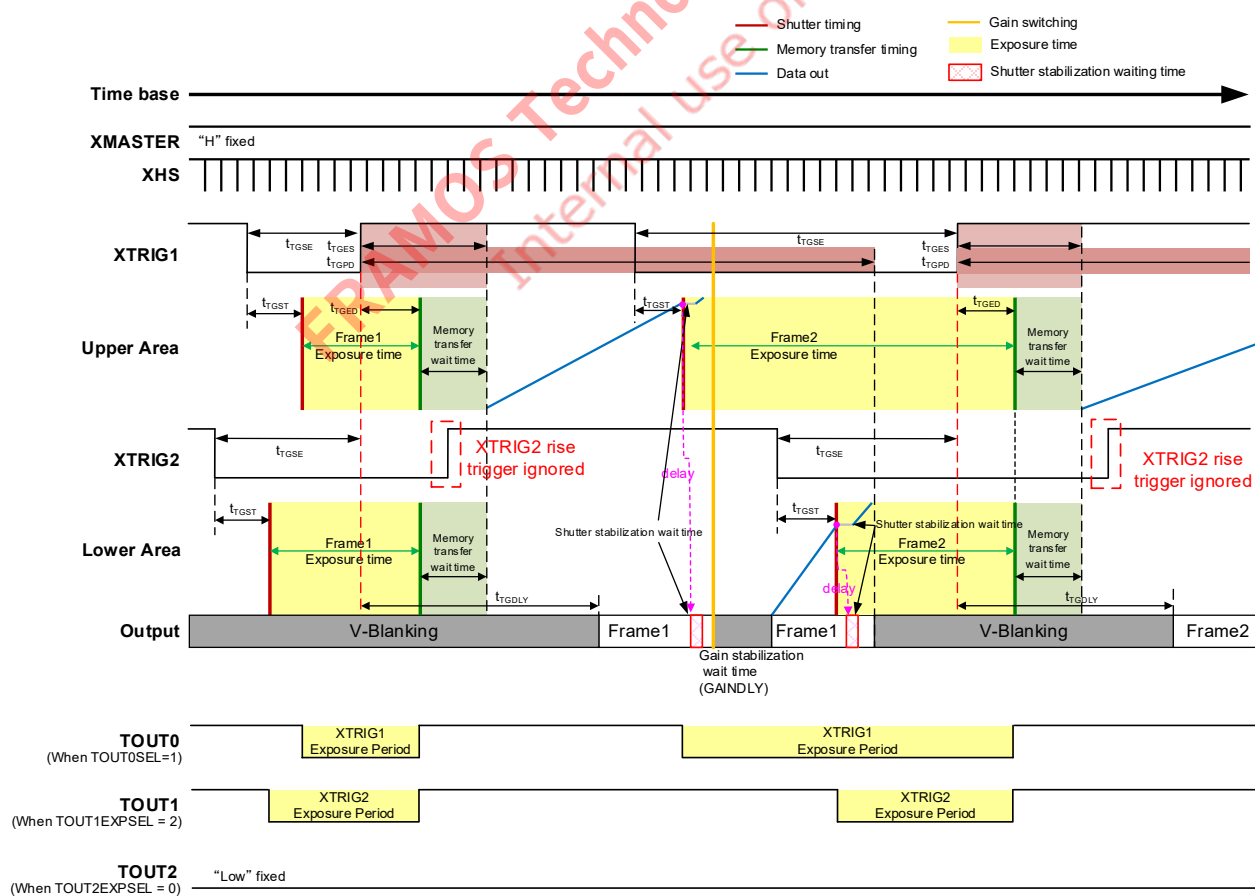
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10. The monitor output for Exposure period

As for this sensor, TOUT0, TOUT1 or TOUT2 can output the status of the exposure control signal in Dual trigger.

10.1. List of registers

Register	Register details			Initial value	Description
	Chip ID	Address (): I ² C	bit		
TOUT0SEL [2:0]	06h	36h (3436h)	[2:0]	0h	TOUT0 pin setting 0h: Low fixed 1h: XTRIG1 Exposure period monitoring 2h: XTRIG2 Exposure period monitoring
TOUT1EXPSEL [2:0]			[5:3]	0h	TOUT1 pin setting 0h: Low fixed 1h: XTRIG1 Exposure period monitoring 2h: XTRIG2 Exposure period monitoring
TOUT2EXPSEL [2:0]		37h (3436h)	[0]	0h	TOUT2 pin setting 0h: Low fixed 1h: XTRIG1 Exposure period monitoring 2h: XTRIG2 Exposure period monitoring



The monitor output for Exposure period of Dual Trigger mode

11. Note

- When transitioning between Normal mode / Trigger mode and Dual Trigger mode, please set via standby mode.
- When changing the Dual trigger area setting (AREAADD [2:0]), please set via standby mode.
- When changing the gain switching setting (AREAGAIN_EN, AREAGAIN_ROISEL [2:0]) of Dual Trigger, please set via standby mode.
- It is possible to change the gain setting values (GAIN [8:0], GAIN 1 [8:0]) during Dual Trigger mode operation.
- Overlap ROI area can't be changed during Dual Trigger mode operation.
- In the Dual Trigger mode, the vertical inverted function can't be used.
- Rise input of XTRIG2 is prohibited before rise input of XTRIG1.
- Dual Trigger mode can't be used with the following function.
 - Multi Exposure Trigger mode
 - Dual ADC (without combination / with built-in combination)
 - Short interval shutter function

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Revision History

Version	Date	Page	Remarks
Rev.0.1	2019 / 05 / 17	–	First Edition
Rev.0.2	2019 / 06 / 20	1, 19	Correction: The function name of the monitor output for exposure time
		20	Correction: The function name of twice exposure
Rev.1.0	2020 / 10 / 27	–	First Edition (Official version of IMX530 series / IMX540 series)
		1	Added Chapter "Product type" Added SLVS 2 ch output in support interface Added the restriction in support interface
		10	Update the table of exposure setting area
		18, 20	Update Parameter list
Rev.2.0	2021 / 3 / 4	1, 2, 4, 11, 12, 20, 24, 25	Added: IMX535 series / IMX545 series / IMX548 series / IMX565 series / IMX568 series
		21, 22	Added: Sequential Trigger Mode (Use gain switching)
Rev.3.0	2021 / 7 / 29	20, 22	Correction: Type name

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